Physical AI SDK

The Low-Resistance Path to AMD Silicon

One Interface · Every Language · CPU · GPU · NPU

AMD
together we advance_

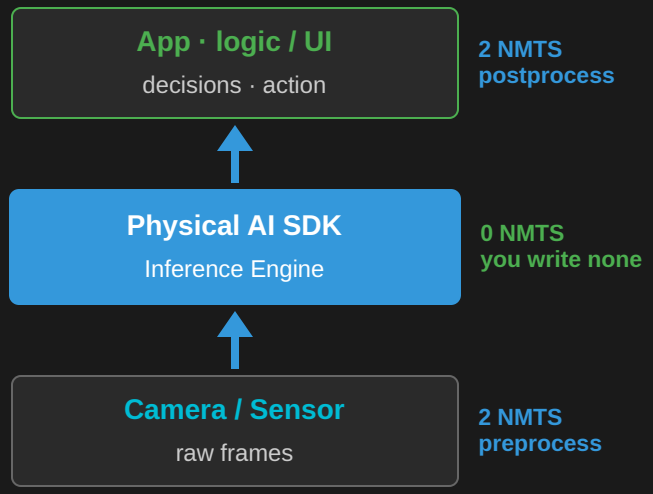
What is Physical AI SDK?

- **Physical AI SDK (PAI SDK)** is a **unified inference engine** that catalyses your product development on AMD hardware.
- It does this by creating a **"Low-Resistance Path"** to AMD silicon.
- A single **unified interface** to run AI workloads across **CPU, GPU, and NPU** on AMD edge hardware.

> **Learn one interface — deploy across every AMD compute device**, from edge CPU to iGPU to NPU.

Physical AI SDK — The Pitch

From Camera to Product — in 4 NMTS



- **2 NMTS in** — capture, hand to the SDK
- **0 NMTS middle** — the SDK *is* the engine
- **2 NMTS out** — act on the results

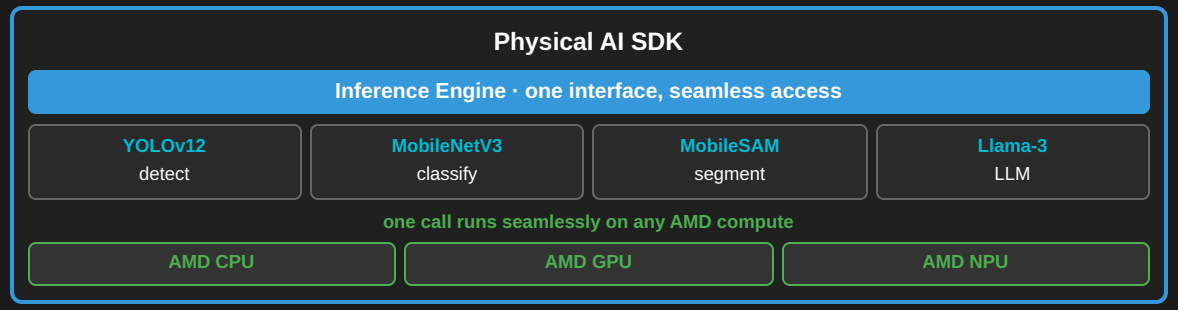
Total effort: 4 NMTS → a working edge app

NMTS = "Not More Than Ten (lines of code)" — one unit of developer effort.

The whole application — in Python

```python
# NMTS 1 · set up the SDK
from physical_ai_sdk.inference_engine import InferenceEngine
engine = InferenceEngine("physical-ai-sdk.local")
yolo   = engine.model("yolov12")
# NMTS 2 · read the camera
frame = Camera.open(0).read()
# NMTS 3 · inference — SDK does everything
dets = yolo.predict(frame)
# NMTS 4 · act on the results
for d in dets:
    draw_box(frame, d.box, d.label)
```

Same call, every language: **Python** **C++** **Kotlin** **JS**



Without PAI SDK — Triton + ROCm + gRPC setup, per-language rewrites, **weeks** to first inference

With PAI SDK — one call, **4 NMTS**, 2 engineers, **half a week** to a stable app on AMD

The Pitch: Physical AI SDK accelerates AI development on AMD Edge-HW — from a quarter to a week

AMD
together we advance_

Same Inference Call — Every Language

Python

```python
from physical_ai_sdk.inference_engine \
    import InferenceEngine
engine = InferenceEngine(
    endpoint="physical-ai-sdk.local")
yolo = engine.model("yolov12")
dets = yolo.predict(frame)
```

C++

```cpp
#include <physical_ai_sdk/inference_engine.hpp>
auto engine = InferenceEngine(
    "physical-ai-sdk.local");
auto yolo = engine.model("yolov12");
auto dets = yolo.predict(frame);
```

Android (Kotlin)

```kotlin
import com.amd.physical_ai_sdk.InferenceEngine
val engine = InferenceEngine(
    endpoint = "physical-ai-sdk.local")
val yolo = engine.model("yolov12")
val dets = yolo.predict(frame)
```

JavaScript / TypeScript

```javascript
import { InferenceEngine }
    from "@amd/physical-ai-sdk";
const engine = new InferenceEngine({
  endpoint: "http://physical-ai-sdk.local:8000" });
const yolo = await engine.model("yolov12");
const dets = await yolo.predict(frame);
```

One mental model, four ecosystems. Same `Detection {label, score, box}` output everywhere.

What Am I Getting in This SDK?

The Engine

- **Unified inference engine** — one interface across **CPU · GPU · NPU**, every language (Python · C++ · Android · JS) and transport (in-place · gRPC · HTTP), **zero-copy on edge**

The Models

- **Optimized model zoo** — AMD-tuned models per task
- **Add a model = drop a config**; parallel · sequential · batched pipelines, all by config

The Proof

- **Accuracy, throughput & PnP benchmarks** on edge for every model — **~130 inf/s on AMD Strix Halo**

The Delivery

- **OBB & Docker** — reproducible, ready-to-ship packaging
- **Co-engineering** with lighthouse customers — cut latency, tune accuracy

> A **low-resistance path** to a **co-engineered, benchmarked, containerized** product on AMD edge hardware.

Pipelines & Models — Config, Not Code

Parallel — describe the topology

```yaml
# pipelines/detect_classify.yaml
name: detect_classify
parallel:
  - model: yolov12       # run
  - model: mobilenetv3   # side by side
```

```
out = engine.model(
    "detect_classify").predict(frame)
out.detections      # from YOLOv12
out.classification  # from MobileNetV3
```

Concurrent — no threads, no GIL, no locks.

Sequential — chain the steps

```yaml
# pipelines/detect_stream.yaml
name: detect_stream
sequential:
  - model: preprocess  # raw → tensor
  - model: yolov12     # tensor → dets
```

Add a model — drop a config

```yaml
# models/my_detector/config.yaml
name: my_detector
task: detect      # auto pre + NMS post
weights: my_detector.onnx
```

Parallel · sequential · batched · load-shared — all a config choice. New models appear on every transport and language automatically.

Physical AI SDK - Structure & Workflow

Three Paths — Pick the One That Fits Your Workflow

Option 1: Zero Steps

Every example ships with a pre-generated `METRICS_TABLE.md`

```
examples/yolov12/METRICS_TABLE.md
```

- CPU, GPU, NPU throughput & latency
- Cross-device comparison table
- Auto-generated from benchmark runs
- **Plan:** expose as a web dashboard

> **Just open the file — numbers are already there**

Option 2: Docker (One Command)

Pre-installed ROCm + RyzenAI environment

```
# Pull & run the container
docker run physical-ai-sdk

# Inside the container:
make benchmark-all-devices metrics
```

- No local installs needed
- Reproducible environment
- Generates `METRICS_TABLE.md` automatically

> **One command to fresh numbers**

Option 3: Manual Setup

Full local installation — 4 steps:

```
# 1. Install ROCm and raizen-ai
./install.sh

# 2. Navigate to example
cd examples/yolov12

# 3. Run benchmarks
make benchmark-all-devices metrics
```

Output: `METRICS_TABLE.md`

> **Most control, most setup**

Easy to Use

Uniform Interface Across Every Example

Make-Based Uniformity

Same commands work in **every** example directory:

```
make benchmark-all-devices metric   # all devices
make benchmark-gpu                  # GPU only
make benchmark-cpu                  # CPU only
make benchmark-npu                  # NPU only
make eval                           # evaluation pipeline
```

- Learn once, use everywhere
- No per-model setup surprises
- Consistent output format (`METRICS_TABLE.md`)

Every Example Includes:

Feature	Description
CPU / GPU / NPU flows	Benchmark each compute device for throughput & latency
Eval pipeline	Sample inputs → sample outputs to visually verify correctness
Guiding documentation	Per-example tutorial to walk you from setup to results

> **Principle:** If you can run one example, you can run them all — same commands, same structure, same outputs.

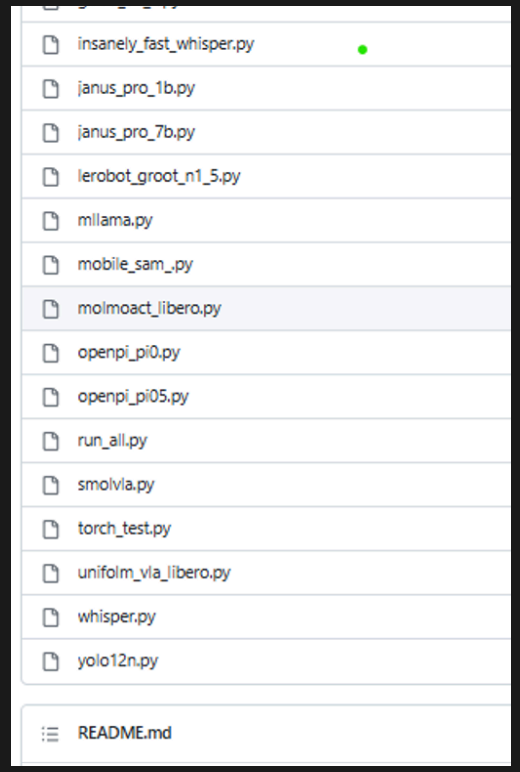
Relation with AIG & Value Add

How PAI SDK builds on — and feeds back into — the AMD AI ecosystem

Contributions — Research Desk → Vertical Software

1. PAVS transforms raw researcher code into production-ready vertical software

What we receive: Research Repo



Flat list of scripts — no structure, no device flows, no Make targets

What we receive: Research Code

```
# rocm-scripts/test/pytorch/yolo12n.py
#........
# .......
# .......
#........
# .......
# .......
recalled = gt_labels & detected_classes
recall = len(recalled) / len(gt_labels)
sanity_pass = recall >= RECALL_THRESHOLD
assert sanity_pass, (
  f"recall {recall:.0%} < {RECALL_THRESHOLD:.0%}")

return lats, {
  "architecture": "CNN",
  "task": "object_detection",
  "num_detections": num_detections,
  "recall": round(recall, 4),
}
if __name__ == "__main__":
  sys.exit(run(run_benchmark))
```

Single script, no structure, no device flows

What we deliver: Vertical Software

```
examples/yolov12/
├── Makefile
├── METRICS_TABLE.md
├── README.md
├── app/
│   ├── routes/
│   ├── services/
│   ├── main.py & config.py
├── scripts/
│   ├── benchmark.py
│   ├── export_onnx.py
│   ├── profile_model.py
│   ├── run_val.py
│   └── update_metrics_table.py
├── tests/
│   ├── test_api.py
│   ├── test_docker.py
│   ├── test_migraphx.py
│   └── test_yolo_workflow.py
├── weights/
├── datasets/
├── requirements.txt
├── requirements_cpu.txt
└── requirements_npu.txt
```

App, scripts, tests, per-device requirements — production-ready

Physical AI & Vertical Software: Researcher's Desk → Vertical Software — same commands, every device, production-ready.

AMD
together we advance_

Contributions(contd.) — Building the AMD AI Ecosystem

2. AIG vs PAVS — Functional Correctness over Operator Validation

Approach	Operator validation	Functional correctness
Inputs	Dummy / synthetic tensors	Real input datasets
Measures	Throughput on isolated ops	End-to-end accuracy + throughput + latency
Pipeline	Op-level checks	Full inference pipeline (pre → infer → post)

> PAVS implements the **end-to-end working pipeline** and ensures the model produces **expected outputs from real input datasets** — not just numbers from dummy inputs.

3. Developer-Friendly Inference Tools & LLM Infra

Profilers + LLM serving stacks — with tutorials for ease of adoption:

Category	Tools
Profilers	Lemonade, NPU AI Analyzer, ROCProfiler
LLM Infra	vLLM (high-throughput serving), Llama.cpp (edge inference), FastFlowLM (LLMs on NPU)

Bridging the **open-source ecosystem** ↔ **AMD HW ecosystem** — making these tools easy to use across the AMD stack.

4. Vertical Application Platform

A platform that **evolves with the client** — solving, fixing, and growing together in targeted domains:

Domain	Focus
Healthcare	Medical imaging, diagnostics
Automotive	Perception, safety systems
Industrial	Inspection, anomaly detection

Contributions(contd.) — AIG Model → Task-Specific

5. Operationalizing AIG Models for Real Applications

From Generic to Task-Specific

Took generic AIG Hub models and extended them with operation enablement, hyperparameter tuning, and backend exploration to reach application-level performance:

Backend	Role
PyTorch	Fallback path — handles missing/broken operators
torch.compile	Additional optimization on top of PyTorch
ONNX	Stable cross-device flow (CPU ↔ GPU ↔ NPU)
MIGraphX	**2× improvement** over torch+ROCm

> **MobileSAM · CenterPoint · YOLOv12 · YOLO26** — integrated into AARTS robotics pipelines, consumed by the robotics team today.

SmolVLA: Tuned for the Task

Latency	48 ms	**30 ms**
ROCm version	—	**7.2.4**
Customized to task	✗	✓

37% latency reduction through algorithmic tuning alone — on ROCm 7.2.4, without the edge-optimized kernels in ROCm 7.13 that give **2× speedup** on other models.

The gains here are purely from **task-specific model and algorithm optimization** — not from a newer ROCm stack.

> **Headroom remains:** upgrading to ROCm 7.13 on top of these algorithmic gains is expected to push latency even lower.

Contributions(contd.) — Ecosystem & Co-Engineering

6. Closing the Loop: Generic Model → Application Pipeline

The Full Journey

Continuing from the previous point — we don't stop at model operationalization:

1. Pick generic model from AIG Hub
2. Make it **task-specific** (tune, operationalize)
3. **Compute accuracy** on real inputs — surface gaps
4. **Communicate gaps back to AIG** for model improvement
5. Wire model into **inference pipelines** — connecting to real applications

> Taking the generic model all the way to working applications — not just benchmarks.

Co-Engineering Opportunities

PAVS × Audio SDK

- Started discussions to explore co-engineering across SDK verticals
- Shared infrastructure, tooling, and pipeline patterns

Rivian (RVT) — Automotive Stack

- Using PAI SDK models to build an **in-car monitoring system** and **voice assistant** on AMD hardware

Google — Gemma on LiteRT

- Running the **Gemma** model on **LiteRT** on AMD hardware

Contributions(contd.) — Feedback Loop Back to AIG

Gaps Found, Communicated, and Tracked

Performance & Deprecation Risks

- **MIGraphX + ONNX is 2× faster than torch.compile** — communicated to AIG and cautioned against deprecating MIGraphX until hipDNN reaches equivalent performance
- **Concat layer broken in YOLOv26** — layer not decomposing correctly, producing garbage outputs; Jiras opened to track and get AIG attention
- **MIGraphX breaks MobileSAM** — missing operator causing failures; AIG informed to investigate the unsupported op

Operator Gaps & Tooling

- **ScatterND not available** in MIGraphX EP or ONNX EP — communicated to AIG for prioritization~(centerpoint)
- **NPU debug tools received** — now able to analyze layer-level outputs that are not producing correct results and provide richer, more precise feedback to AIG
- **Unified NPU + GPU dependency request** — GPU stack requires NumPy <2, NPU stack requires NumPy 2+; the version conflict forces separate virtual environments today. Communicated to AIG to resolve this so NPU and GPU can share a single application-level environment

> **PAVS acts as the functional correctness signal for AIG** — we find what breaks on real inputs and real pipelines, and feed that back so the stack improves.

Contributions(contd.) — PAI SDK Platform & Edge Enablement

Infrastructure: Installer, Edge HW, and LLM Stack

Installation & Edge Hardware

- **Single installer/uninstaller** for ROCm + RyzenAI in PAI SDK — one command to set up or tear down the full stack
- Extended and maintained ROCm + RAI support for **edge HW** — required targeted patches for hardware-specific gaps
- **Yocto & LTS support** — enabling model deployment on embedded edge devices

LLM Infrastructure on AMD Stack

- Integrated **stable vLLM** — high-throughput serving with quantized LLM variants on iGPU
- Integrating **FLM framework** — running LLMs on NPU
- Bridging the open-source LLM ecosystem onto the AMD edge stack

> Full-stack edge enablement: installer → hardware patches → quantized LLM serving → NPU inference → Yocto deployment.

Pain Points & Solution

Current Challenges on Edge Devices — and How We've Addressed Them

Challenges

ROCm Installation Stability

- ROCm install on edge devices is **not yet stable**
- Driver/firmware mismatches on some hardware revisions

Reboot During Installation

- ROCm installation requires a **system reboot** mid-process
- Breaks unattended / CI-driven installs

Sudo / Root Access Required

- Both ROCm and RyzenAI installers need **sudo privileges**

Solution: Dockerized PAI SDK

We have **dockerized the full setup** — ROCm + RyzenAI pre-installed, pre-configured, and ready to use out of the box:

```
# Pull and run — no installs, no reboot, no sudo
docker run physical-ai-sdk

# Start benchmarking immediately
make benchmark-all-devices metrics
```

- No local ROCm install required
- No reboot, no sudo on host
- Reproducible environment across machines
- **Ready to deliver to users today**

> **Docker sidesteps all three pain points** — users get the full PAI SDK on the fly, without touching the host system.

AMD
together we advance_

SW Stack (For Reference)